

The Angular Observer as a Manifold Recognizer

Reading Emergent Geometry from Hypergraph Rewriting

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Abstract

If space is a discrete network evolving under local rewrite rules, physics needs an instrument that zooms out: given only nodes and edges, decide whether a macro-geometry has emerged, and if so, *which one*. The companion paper [1] established the measuring tape—the angular coordinate of the commute-weighted low-mode Laplacian embedding preserves graph geodesics where the raw embedding degenerates into local density. This paper turns the measuring tape into a *recognizer*: a battery of five instruments (density irregularity, dimension consensus, angular fidelity with controls, a shape battery of flat-embedding stress / sphere-fit residual / eccentricity homogeneity, and a low-spectrum multiplet signature scored against closed-manifold templates) that identifies the manifold behind a graph, or reports that there is none. The battery validates on ground truth: a sampled flat torus is classified as a flat torus and a sampled sphere as S^2 , from spectra and angular distances alone. Pointed at five Wolfram-model rewrite rules, it returns graded, interpretable verdicts: no rule grows a closed manifold (all are bounded patches with an edge, eccentricity spread 0.39–0.65 against ≈ 0.10 for the closed controls); one rule is, at every growth stage and seed tested, an almost perfect *interval* (spectrum ratios 1, 3.8, 8.7, 14.8 against the 1, 4, 9, 16 template); two are unambiguously two-dimensional bounded patches whose finer symmetry class is less stable than their dimension; one is genuinely fractal ($d \approx 1.4$); and one is a ≈ 3.4 -dimensional space matching no manifold template. Two findings are new. First, under a growth sweep (750–12,000 rewrite events), *dimension emerges before shape*: a rule’s measured dimensionality locks in at the earliest stages and is stable throughout, while its finer symmetry class (patch vs. sphere vs. torus) can keep wandering—dimensionality is the robust emergent invariant. Second, on emergent substrates the Coifman–Lafon $\alpha=1$ density correction *lowers* angular fidelity even as it improves sampled manifolds—on a graph with no latent manifold, density is structure, not sampling noise.

1 Introduction

In discrete-spacetime programs—causal sets, causal dynamical triangulations, and the Wolfram model [7, 8]—the universe is a graph or hypergraph and continuum geometry is supposed to emerge at scale. That supposition needs an instrument. To say a hypergraph “is” an emergent 2-manifold is to claim that some coarse-graining of its combinatorics converges to the metric geometry of a surface; the claim is empty without a way to compute the coarse-graining and check the convergence.

Spectral embeddings are the natural candidate instrument, and they carry a known pathology that is fatal in exactly this setting: on large geometric graphs the commute (resistance) distance degenerates to a function of local degrees [2], and rewrite rules—localized, uncoordinated—produce precisely the irregular node densities that trigger the degeneracy. The companion paper [1] showed

that the degeneracy is confined to the *radial* coordinate of the embedding: row-normalization (the Ng–Jordan–Weiss step [3]) deletes the density factor, and the surviving *angular* coordinate preserves graph geodesics—with a conditional transfer theorem, an unconditional instance on the flat torus, and rank fidelity flat in mode budget and graph size across substrate families.

That result is a measuring tape: it certifies that angular distances track geodesics. This paper asks the next question. Given the tape, can we read off *which* geometry a rewrite rule grows—recover smooth, recognizable manifolds where they exist, and detect the spaces that are something else entirely? We assemble a recognition battery (§2), validate it on manifolds with known ground truth (§3), and point it at five Wolfram-model rules (§4). Two phenomena beyond recognition emerge: the verdicts *move* as substrates grow (§5), and the standard density-correction of manifold learning is the wrong move on emergent substrates (§6).

2 The recognition battery

Let G be the clique-expanded graph of an evolved hypergraph (largest component, cropped to $n \leq 2000$), L_{sym} its symmetric-normalized Laplacian with eigenpairs (λ_k, v_k) . The angular embedding is the row-normalized commute-weighted map $\hat{u}_i = \Psi_i / \|\Psi_i\|$, $\Psi_i = (v_k(i) / \sqrt{\lambda_k})_{k=1}^m$, $m = 20$, exactly as in [1]; angular distances are chordal, geodesics are hop distances on 150 anchors. The battery has five instruments.

(1) Density irregularity. The degree coefficient of variation $\text{CV}(k_i)$: the driver of commute degeneracy [2], and the regime rewrite rules inhabit.

(2) Dimension consensus. Ball-growth, spectral (heat-trace), and effective-rank estimators; their mean is the measured intrinsic dimension d and their spread the manifold-grade criterion, as in [1].

(3) Fidelity with controls. Spearman angle- ρ between angular distances and geodesics; magnitude-only ρ ; an equal-size random-mode control (must be ≈ 0); and the Coifman–Lafon $\alpha=1$ density-normalized angle [4] for §6.

(4) Shape battery. Three coordinate-free discriminators of the angular geometry D_{ang} on the anchor set: *flat-stress*, the residual stress of the best rank-2 (and rank-3) classical MDS of D_{ang} —a flat patch embeds at low stress, a closed or curved geometry cannot; *sphere fit*, the relative RMS radial residual of the best-fit sphere to the 3D MDS configuration; *homogeneity*, the spread $(\max - \min)/\text{mean}$ of anchor eccentricities—closed homogeneous manifolds are narrow, bounded patches are broad.

(5) Multiplet signature. The low-spectrum ratios λ_k/λ_1 , $k = 1, \dots, 8$, scored (RMS log-distance) against four templates: flat 2-torus $(1, 1, 1, 1, 2, 2, 2, 2)$, sphere S^2 $(1, 1, 1, 3, 3, 3, 3, 3)$, Neumann square patch $(1, 1, 2, 4, 4, 5, 5, 8)$, and interval $(1, 4, 9, 16, \dots)$. Multiplicity structure is a fingerprint of symmetry and topology that survives discretization; for random geometric graphs the low normalized-Laplacian spectrum converges to the Laplace–Beltrami spectrum [5, 6], so the templates are the right asymptotic targets.

3 Validation on known manifolds

The battery must first recognize geometry we can check. On a flat 2-torus random geometric graph ($n = 2000$) and a sphere S^2 graph ($n = 2000$): the torus is classified *flat 2-torus* (signature distance 0.20 vs 0.34 for the sphere template) and the sphere is classified *sphere* (0.17 vs 0.56 for the torus template) with the smallest sphere-fit residual of any substrate (0.10); both show the closed-manifold profile the shape battery predicts—high flat-stress (0.54, 0.48: neither flattens into

$\mathbb{R}^2$) and near-uniform eccentricities (spread 0.11, 0.09). Angle- ρ is 0.90 and 0.96; magnitude and random controls are ≤ 0.04 . The instrument can tell a torus from a sphere from a patch using spectra and angular distances alone.

4 Five rules, five verdicts

We point the battery at the five hypergraph-rewrite rules of the companion paper’s manifold library (arity-3, grown to the full 3000-edge budget with budget-scaled generation cap, seed 1; seed stability in §4.1).

rule	n	deg CV	d (spread)	angle- ρ	stress ₂	ecc. spr.	verdict
torus control	2000	0.25	1.93 (0.09)	0.90	0.54	0.11	flat 2-torus
sphere control	2000	0.26	1.83 (0.12)	0.96	0.48	0.09	sphere S^2
R_2D	2000	0.28	1.05 (0.29)	0.93	0.21	0.64	interval (a curve)
R_frac	2000	0.06	1.39 (0.08)	0.89	0.33	0.65	fractal filament web
evolved-2D	2000	1.42	2.03 (0.31)	0.73	0.43	0.55	ragged 2D patch
R_3D	2000	0.18	2.12 (0.34)	0.67	0.57	0.50	2D, torus-leaning
R_SR	2000	0.90	3.36 (0.28)	0.55	0.49	0.39	no template ($\approx 3.4D$)

Four readings. **(i) No rule grows a closed manifold.** Every emergent substrate has eccentricity spread 0.39–0.65 against ≈ 0.10 for the closed controls: these are *bounded patches grown outward from a seed*, geometry with an edge—as a finite rewriting history should produce. **(ii) The recognizable cases are genuinely recognizable.** R_2D is identified as an *interval* with near-exact spectrum (1, 3.8, 8.7, 14.8, 24.0 against 1, 4, 9, 16, 25; signature distance 0.048); R_3D is unambiguously two-dimensional—all 2D templates score far below interval (0.28–0.68 vs 2.32), flat-torus-leaning at this budget, though §5 shows the winning 2D template wanders with growth, so “2D” is the claim and the finer symmetry class is not. **(iii) The unexpected spaces are flagged, not force-fitted.** R_frac is genuinely fractal ($d \approx 1.4$, filament clusters); R_SR is a ≈ 3.4 -dimensional space whose spectrum matches no manifold template. **(iv) Dimension gating shows up cross-substrate.** The two quasi-1D rules have the highest magnitude- ρ (0.29, 0.47): the radius stays geodesic-informative exactly where the radial-degeneracy corollary of [1] does not apply. Random-mode controls are ≤ 0.02 everywhere.

4.1 Seed stability

Each verdict above is one growth instance; rewrite rules are nondeterministic in general, so verdicts must survive the rewriter seed. Re-running the full battery at seeds 1–4 (full 3000-edge budget):

R_2D: stable, 4/4 interval ($d = 1.05 \pm 0.01$, angle- $\rho = 0.93 \pm 0.00$)—the curve is a property of the rule. **R_frac: dimension rock-stable** ($d = 1.39 \pm 0.00$ across seeds), with the template tipping once (3/4 interval, 1/4 square patch): a fractal filament web genuinely sits between the two templates. **R_3D: growth is deterministic**—all four seeds produce byte-identical spectra ($d = 2.12$, angle- $\rho = 0.67$), so this rule evidently admits a single applicable rewrite per step; seed replication is vacuous for it and the growth axis of §5 is the meaningful robustness check. **evolved-2D: genuinely seed-dependent:** $d = 1.87 \pm 0.14$ but angle- ρ ranges 0.38–0.89 and the verdict scatters (patch/torus/interval)—this rule does not reliably grow the same geometry, and one seed grows a substrate whose angular fidelity collapses. **R_SR: 4/4 non-manifold** ($d = 3.39 \pm 0.11$): the winning template flips per seed with uniformly weak scores (0.25–0.55), which is what “matches none of these” should look like under replication. The meta-pattern of §5 repeats across seeds:

2D MDS of ANGULAR distances — what the angular observer sees

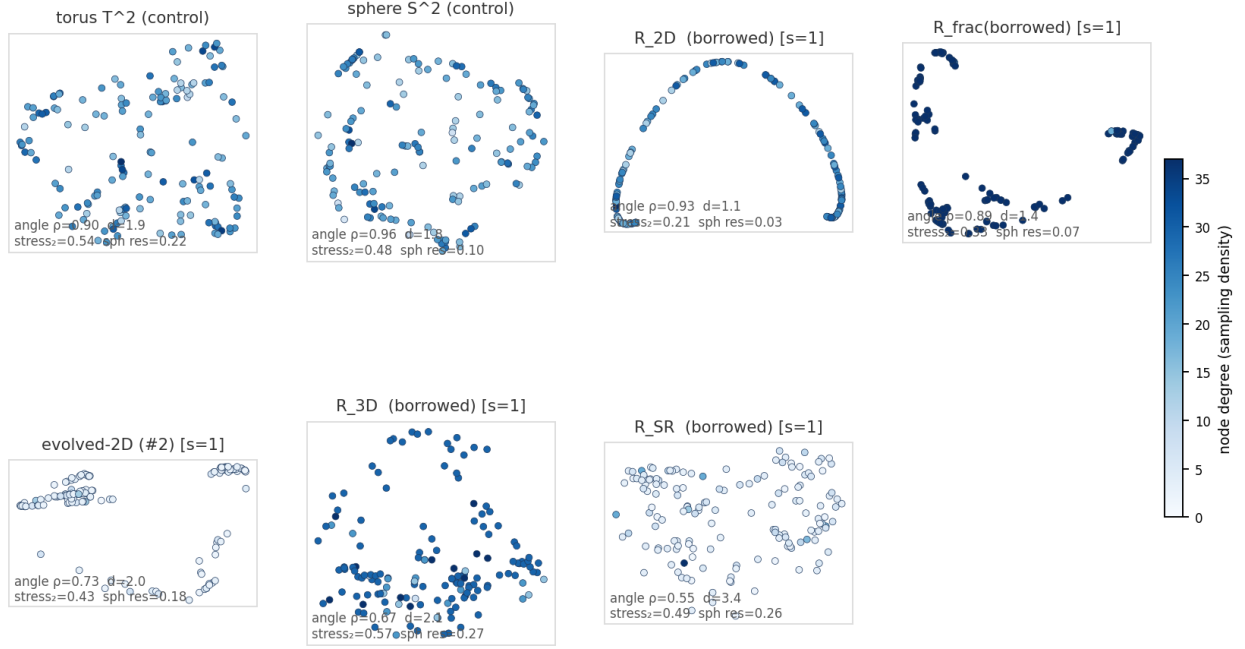


Figure 1: What the angular observer sees: 2D MDS of the *angular* distance matrix per substrate, points shaded by node degree (sampling density). The closed controls fill the plane (they cannot be flattened); R_2D collapses onto a curve; R_3D and evolved-2D are bounded 2D patches; R_SR is a diffuse high-dimensional cloud.

dimension is the stable invariant (± 0.14 at worst); the finer symmetry class is stable only where the geometry actually is one of the model spaces.

5 Dimension emerges before shape

Is “R_2D is a curve” a fact about the rule or about the growth stage? We sweep the rewrite budget over 750–12,000 edges (generation cap scaled with the budget; beyond 3000 edges the recognizer reads a fixed 2000-node geodesic-ball window of the larger substrate, i.e. a bounded observer probing a growing universe) and re-run the recognizer at each stage (Fig. 3). Three growth behaviours appear, none the naive “template crossing” one might expect:

R_2D is a curve, definitively. The interval signature is pinned at distance 0.02–0.05 at *every* stage, with $d = 1.04$ – 1.06 and angle- $\rho = 0.93$ – 0.95 throughout; the interior of the large substrate is as one-dimensional as its early stages. The verdict of §4 was about the rule, not the stage.

evolved-2D is a stable patch. Square-patch wins at all five budgets ($d = 1.94$ – 2.05): a robust, reproducible-under-growth identity.

R_3D: dimension locks first, shape does not. The measured dimension sits at $d \approx 1.9$ – 2.2 from the earliest stage, and all three 2D templates score in a tight band (0.24–0.68) far below interval (1.85–2.32)—but the *winning* 2D template wanders: square patch (750) $\rightarrow$ sphere (1500)

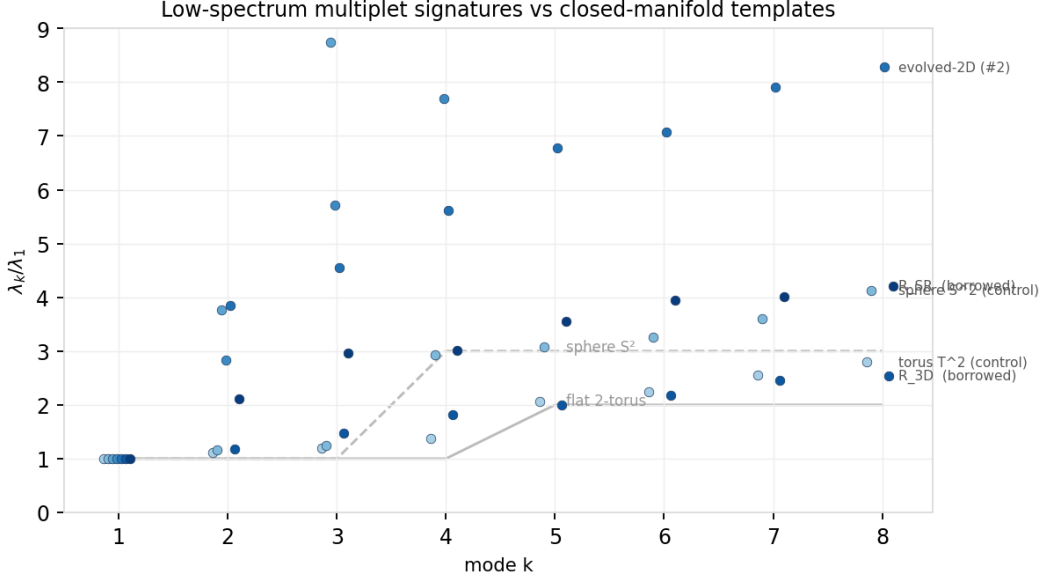


Figure 2: Low-spectrum multiplet signatures λ_k/λ_1 against closed-manifold templates. The controls hug their templates (torus the solid line, sphere the dashed); the emergent rules climb far faster—no multiplet structure, hence no closed symmetry—which is exactly how the battery separates patches and non-manifolds from closed geometry.

→ flat torus (3000) → sphere (6000) → square patch (12,000). The refined claim this supports: *dimensionality is the robust emergent invariant; symmetry class is not*—at these scales R_3D is unambiguously a 2D geometry that is not yet (or not ever) any of the clean model spaces. Its angular fidelity also dips and recovers across the sweep ($0.80 \rightarrow 0.59 \rightarrow 0.73$), consistent with the fixed-budget observer effect documented in the companion paper’s continuum section: a fixed $m = 20$ window resolves a shrinking fraction of a growing substrate.

6 Density is structure: the $\alpha=1$ dissociation

On sampled manifolds, non-uniform node density is *sampling noise*: the graph Laplacian converges to a density-weighted operator, and the Coifman–Lafon $\alpha=1$ normalization removes the weight, demonstrably restoring angular fidelity under strong contrast [4, 1]. The recognizer reproduces that on the controls (S^2 : $0.96 \rightarrow 0.98$; torus: $0.90 \rightarrow 0.91$). On the emergent substrates the same correction moves fidelity the *other way*, and the seed sweep (§4.1, 20 instances) shows the effect is *graded in density irregularity*: for the low-irregularity rules ($CV \leq 0.3$: R_2D, R_frac, R_3D) the $\alpha=1$ shift is -0.001 to $+0.011$ —indistinguishable from zero—while for the high-irregularity rules it is consistently negative: R_SR ($CV \approx 0.9$) loses 0.03–0.06 at every seed, and evolved-2D ($CV \approx 1.4$) loses up to 0.04 in three of four seeds. The reading: on a graph with no latent manifold underneath, the density profile is not noise over a ground truth—it *is* part of the substrate’s structure, and the more structured the density, the more signal the correction removes. This gives the density coordinate an unexpected second life: for sampled geometry it is a nuisance parameter, for emergent geometry it is data, and the sign of the $\alpha=1$ response is itself a diagnostic for which situation one is in.

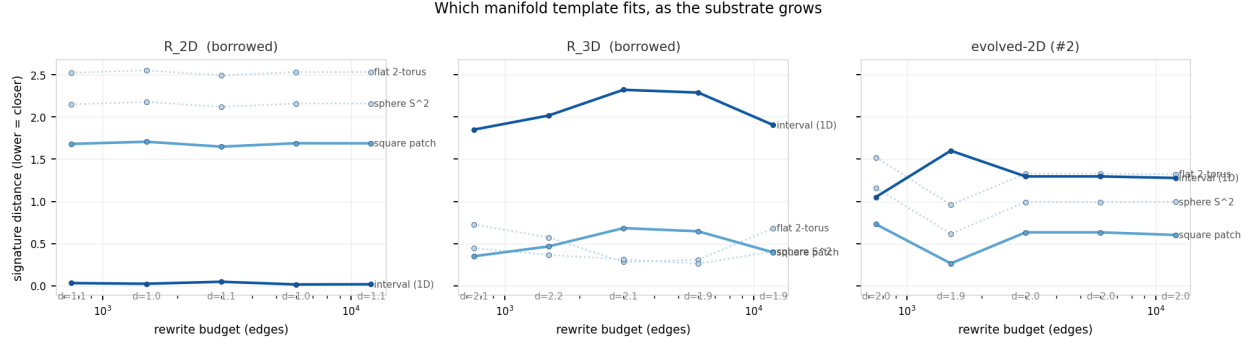


Figure 3: Which manifold template fits, as the substrate grows: multiplet signature distance per template (lower = closer) across rewrite budgets, measured dimension annotated per stage. R_2D (left) is an interval at every stage; evolved-2D (right) is a stable square patch; R_3D (middle) locks its *dimension* at ≈ 2 immediately while the winning 2D template wanders—dimension emerges before shape.

7 Limitations and falsifiers

Seed replication is done (§4.1: two rules fully stable, one deterministic, one tipping between adjacent templates, one genuinely seed-dependent), but the growth sweep remains single-seed per stage; hop-count geodesics; $n \leq 2000$ dense-eigensolver scale; four templates, so “matches none” means none-of-these—hyperbolic, higher-genus, and product templates are the obvious extensions; the growth sweep is a budget sweep, not a fixed-rule time series at scale. Each is mechanical to lift, none is conceptual. The recognizer is falsifiable in use: it must keep classifying held-out sampled manifolds correctly (cylinders, genus-2 surfaces, T^3), and its emergent-substrate verdicts must be stable under seeds and reproduce under budget for the claims here to stand.

8 Outlook

The bridge claim, stated with its bounds: for a universe built purely of nodes and edges, the angular coordinate of the low-mode spectral embedding is a mathematically controlled zoom-out—the companion paper supplies the guarantees, and this paper shows the zoom-out is *legible*: it does not merely score geodesic fidelity, it reads off dimension, boundedness, curvature class, and symmetry, and it declines to hallucinate a manifold where none exists. In observer-theoretic terms [9], the recognizer is a concrete, falsifiable candidate for the coarse-graining a bounded observer performs; nothing here requires that reading, and the strongest physics-adjacent findings—bounded patches rather than closed universes, dimension arriving under growth, density as structure—stand as graph-theoretic facts about rewrite systems.

Reproducibility. experiments/manifold-recovery/ in the companion repository (<https://github.com/ahb-sjsu/the-angular-observer>) contains the recognizer (`recover.py`), the growth sweep (`growth_sweep.py`), committed result JSONs, and the figure scripts; all runs are seeded, CPU-only, NumPy/SciPy.

References

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